

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Introduction to Bluetooth

Bluetooth is a short-range wireless communication technology designed to connect devices such as:

- Mobile Phones
- Laptops
- Printers
- Cameras
- Keyboards and Mice
- Smart Home Devices

Characteristics:

- Low power consumption
- Low cost
- Short range communication
- Operates in the 2.4 GHz ISM band

Applications of Bluetooth

Bluetooth is widely used in:

- Wireless mouse and keyboards
- Wireless headphones and speakers
- Smart watches
- Medical monitoring devices
- Home automation systems
- Device synchronization

Examples:

- Fitness trackers
- Smart locks
- Smart lighting systems

Bluetooth Overview

Bluetooth is based on:

IEEE 802.15

Defines:

- Wireless Personal Area Network (WPAN)
- Room-sized communication network
- Ad hoc networking

Typical Range:

10 m

Typical Data Rate:

1 Mbps

Bluetooth Network Architecture

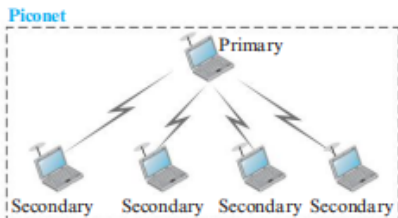
Bluetooth defines two network structures:

- 1 Piconet
- 2 Scatternet



These structures enable flexible communication among devices.

Figure 6.19 *Piconet*



A Bluetooth network is called a Piconet.

Characteristics:

- One Primary device
- Up to 7 Active Secondary devices
- Additional devices can remain parked
- All devices synchronize with Primary

Piconet Communication

Communication Modes:

- One-to-One
- One-to-Many

Important Points:

- Only one Primary exists
- Secondaries communicate through Primary
- Secondary-to-Secondary communication is indirect



Parked State

Bluetooth supports additional devices using parked mode.

Parked Device:

- Synchronized with Primary
- Cannot actively communicate
- Consumes less resources

To activate a parked device:

- One active device must move to parked state

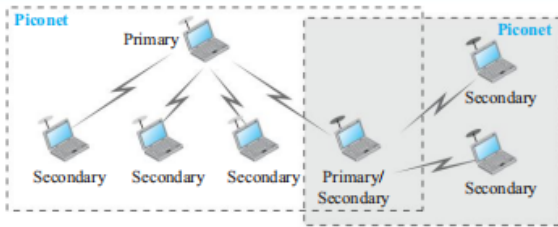
Multiple piconets can be interconnected to form a:

Scatternet

Characteristics:

- Multiple piconets interconnected
- A device may belong to multiple piconets
- A Secondary in one piconet may act as Primary in another

Figure 6.20 *Scatternet*



Advantages of Scatternet

Benefits:

- Increased coverage
- Larger network size
- Better scalability
- Improved connectivity



Bluetooth Protocol Stack

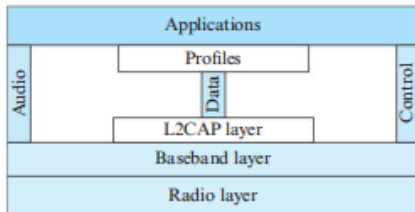
Bluetooth uses a layered architecture.

Main Layers:

- 1 L2CAP Layer
- 2 Baseband Layer
- 3 Radio Layer



Figure 6.21 *Bluetooth layers*



L2CAP:

Logical Link Control and Adaptation Protocol

Equivalent to LLC layer in LANs.

Functions:

- Multiplexing
- Segmentation
- Reassembly
- QoS Support
- Group Management



L2CAP Packet Format

Fields:

- Length Field (16 bits)
- Channel Identifier (CID)
- Data



Maximum Data Size:

65,535 bytes

Figure 6.22 *L2CAP data packet format*



Multiplexing

- Supports multiple upper-layer protocols.

Segmentation and Reassembly

- Divides large packets into smaller segments.
- Reassembles at destination.

QoS Support

- Provides Quality of Service.
- Default mode is Best Effort.

L2CAP supports logical groups.

Advantages:

- Similar to multicasting
- One-to-many communication
- Efficient resource usage

Baseband Layer

Equivalent to MAC Sublayer.

Functions:

- Access control
- Link management
- Frame transmission
- Error control

Access Method:

TDD – TDMA

TDD-TDMA:

Time Division Duplex TDMA

Characteristics:

- Half-duplex communication
- Separate slots for transmission
- Uses time slots of $625 \mu s$

Single Secondary Communication

Slot Allocation:

- Primary uses Even Slots
- Secondary uses Odd Slots

Example:



Primary:

0, 2, 4

Secondary:

1, 3, 5

Figure 6.23 Single-secondary communication

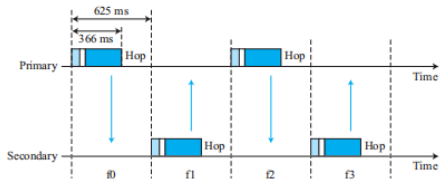
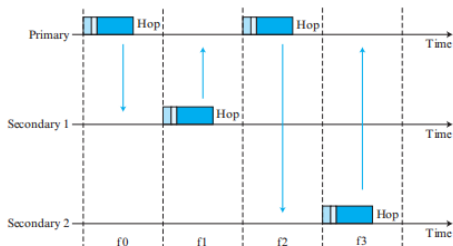


Figure 6.24 Multiple-secondary communication



Multiple Secondary Communication

When multiple secondaries exist:

- Primary uses all even slots.
- Addressed secondary replies in next odd slot.
- Other secondaries remain silent.

Figure 6.23 *Single-secondary communication*

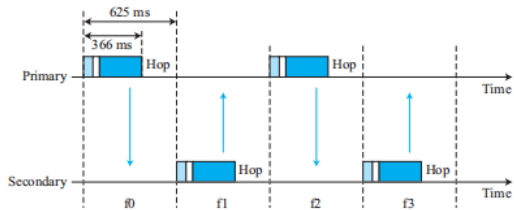
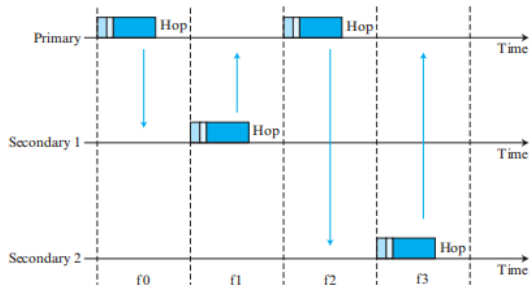


Figure 6.24 *Multiple-secondary communication*



Bluetooth Link Types

Bluetooth supports:

- 1 SCO Link
- 2 ACL Link



SCO:

Synchronous Connection Oriented

Characteristics:

- Delay-sensitive
- Real-time audio communication
- Fixed slot reservation
- No retransmission

Data Rate:

64 kbps

ACL:

Asynchronous Connectionless Link

Characteristics:

- Error-free delivery preferred
- Retransmission supported
- Suitable for data transfer

Maximum Data Rate:

721 kbps

SCO vs ACL

SCO	ACL
Real-time Audio	Data Communication
No Retransmission	Retransmission Allowed
Delay Sensitive	Reliability Sensitive
64 kbps	721 kbps

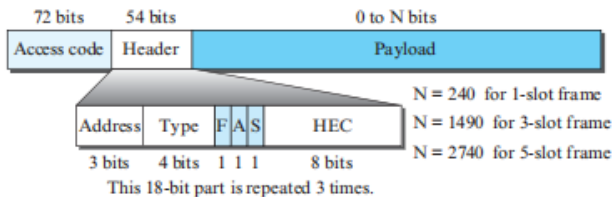
Bluetooth Frame Format

Frame Types:

- One Slot Frame
- Three Slot Frame
- Five Slot Frame



Figure 6.25 *Frame format types*



Bluetooth Frame Fields

Major Fields:

- 1 Access Code (72 bits)
- 2 Header (54 bits)
- 3 Payload (0–2740 bits)



Header Fields

Header Components:

- Address (3 bits)
- Type (4 bits)
- Flow Control (F)
- Acknowledgment (A)
- Sequence Number (S)
- HEC

Uses Stop-and-Wait ARQ.

Bluetooth Radio Layer

Equivalent to Physical Layer.

Characteristics:

- Low power operation
- Short range (10 m)
- 2.4 GHz ISM Band



Bluetooth Frequency Allocation

Frequency Band:

2.4 GHz

Channels:



Each Channel:

1 MHz

Bluetooth uses:

FHSS

(Frequency Hopping Spread Spectrum)

Characteristics:

- 1600 hops per second
- Dwell Time = $625\ \mu s$
- Reduced interference
- Enhanced security

Bluetooth Modulation

Modulation Technique:

GFSK

Gaussian Frequency Shift Keying

Bit Representation:

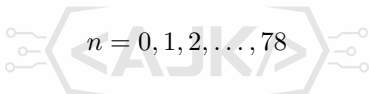
- Bit 1 → Frequency above carrier
- Bit 0 → Frequency below carrier

Carrier Frequency Formula

Carrier Frequency:

$$f_c = 2402 + n$$

where


$$n = 0, 1, 2, \dots, 78$$

Examples:

- Channel 0 = 2402 MHz
- Channel 1 = 2403 MHz
- Channel 78 = 2480 MHz

Bluetooth Summary

- Bluetooth follows IEEE 802.15.
- Uses Piconets and Scatternets.
- Supports up to 8 active devices in a Piconet.
- Uses TDD-TDMA access.
- Supports SCO and ACL links.
- Operates in 2.4 GHz ISM Band.
- Uses FHSS and GFSK.
- Provides low-power short-range communication.

Exam Focus

Piconet, Scatternet, L2CAP, TDD-TDMA, SCO, ACL, Bluetooth Frame Format, FHSS, GFSK.

Introduction to WiMAX

WiMAX stands for

Worldwide Interoperability for Microwave Access

IEEE Standards:

- IEEE 802.16 (Fixed WiMAX)
- IEEE 802.16e (Mobile WiMAX)

Purpose:

- Broadband wireless access
- Alternative to DSL
- Alternative to Cable Modem
- Alternative to Cellular Data Services

Need for WiMAX

WiMAX was developed to solve the:

"Last Mile Connectivity Problem"

Challenges:

- Expensive cable deployment
- Difficult terrain
- Rural broadband access
- Long-distance wireless connectivity

Solution:

- High-speed wireless broadband
- Large coverage area
- Cost-effective deployment

WiMAX vs WiFi

Feature	WiFi	WiMAX
IEEE Standard	802.11	802.16
Coverage	100 yards	6 miles
Mobility	Limited	High
QoS Support	Basic	Advanced
Security	Moderate	Strong
Application	LAN	MAN/Broadband

WiMAX Architecture

Main Components:

- 1 Base Station (BS)
- 2 Subscriber Station (SS)
- 3 Portable Units



Communication Flow

Subscriber ↔ Base Station ↔ Internet

WiMAX Architecture Diagram

Insert WiMAX Architecture Figure Here

Components:

- Internet Backbone
- WiMAX Base Station
- Subscriber Stations
- Portable Devices



WiMAX Base Station

A Base Station consists of:

- Radio Transmitter
- Radio Receiver
- Antenna System

Operating Frequency:



$2\text{ GHz} - 11\text{ GHz}$

WiMAX uses:

Software Defined Radio (SDR)

Software Defined Radio (SDR)

SDR performs radio functions using software.

Advantages:

- Flexibility
- Easy upgrades
- Multiple protocol support
- Reduced hardware complexity



Types of WiMAX Antennas

WiMAX uses three antenna types:

- 1 Omni-directional Antenna
- 2 Sector Antenna
- 3 Panel Antenna

Purpose:

- Increase coverage
- Improve throughput
- Reduce interference



Omni-Directional Antenna

Characteristics:

- Radiates equally in all directions
- 360° coverage
- Suitable for general broadcasting

Applications:

- City-wide coverage
- Public hotspots

Sector Antenna

Characteristics:

- Covers specific angular sectors
- Higher gain
- Reduced interference

Typical Coverage:

60°, 120°, 180°

Panel Antenna

Characteristics:

- Highly directional
- Focused transmission
- Long-distance communication

Applications:

- Point-to-point communication
- Backhaul links

Adaptive Antenna System (AAS)

WiMAX uses:

Adaptive Antenna System (AAS)

Functions:

- Beam Steering
- Focused Transmission
- Focused Reception
- Reduced Interference



Benefits:

- Higher throughput
- Better signal quality

Beam Steering Concept

Traditional Antenna:

- Broadcasts energy everywhere

AAS Antenna:

- Directs energy toward receiver
- Improves signal strength
- Conserves power

WiMAX uses:

OFDMA

Orthogonal Frequency Division Multiple Access

Advantages:

- Multiple users simultaneously
- High spectral efficiency
- Reduced interference
- Better bandwidth utilization

MIMO:

Multiple Input Multiple Output

Uses:

- Multiple transmit antennas
- Multiple receive antennas

Benefits:

- Increased throughput
- Improved reliability
- Better signal quality

Customer Premises Equipment (CPE)

Types:

- 1 Indoor Unit
- 2 Outdoor Unit



Characteristics:

- Similar to DSL modem
- Easy installation
- Self-installed
- Requires proximity to Base Station



Outdoor Subscriber Unit

Characteristics:

- Similar to satellite dish
- Higher signal gain
- Longer communication range
- Professional installation required

Portable WiMAX Devices include:

- Mobile phones
- Laptops
- Tablets
- Gaming devices
- MP3 players
- Embedded IoT devices



Supports:

Mobile WiMAX

WiMAX Data-Link Layer

WiMAX MAC uses:

Scheduling Based Access

Unlike WiFi:

- No continuous contention
- Better fairness
- Better QoS

WiFi vs WiMAX MAC

Feature	WiFi	WiMAX
Access Method	Contention Based	Scheduling Based
QoS	Limited	Strong
Efficiency	Moderate	High
Scalability	Medium	High

Scheduling Algorithm

Process:

- 1 Subscriber requests access.
- 2 Base Station grants slot.
- 3 Dedicated transmission slot allocated.
- 4 Data transmission occurs.

Advantages:

- Predictable performance
- Better bandwidth utilization
- Reduced collisions

IEEE 802.16e Features:

- Frequency Range: 2–11 GHz
- SOFDMA
- MIMO
- Full Mobility Support



SOFDMA:

Scalable Orthogonal Frequency Division Multiple Access

Benefits:

- Flexible bandwidth allocation
- Efficient spectrum usage
- Supports many users
- Improved throughput

Applications of WiMAX

WiMAX can be used for:

- Broadband Internet Access
- Rural Connectivity
- Last-Mile Connectivity
- Cellular Backhaul
- Enterprise Networking
- Disaster Recovery Networks

Advantages of WiMAX

- Long-range communication
- High throughput
- Strong QoS support
- Enhanced security
- Mobility support
- Cost-effective deployment



Limitations of WiMAX

- Infrastructure cost
- Spectrum licensing issues
- Weather effects
- Competition from LTE and 5G

WiMAX Summary

- IEEE 802.16 standard.
- Provides broadband wireless access.
- Covers up to 6 miles.
- Uses SDR, OFDMA, AAS and MIMO.
- Scheduling-based MAC.
- Supports fixed and mobile wireless communication.
- Designed for last-mile connectivity.

Exam Focus

WiMAX Architecture, Base Station, AAS, OFDMA, MIMO, Scheduling MAC, SOFDMA, Applications.